

Convex Combination

BM25 score

mDPR score

$$s_{\text{CC}} = \alpha \cdot \tilde{s}_{\text{Dense}} + (1 - \alpha) \cdot \tilde{s}_{\text{BM25}}$$

Final ranking

Reciprocal Rank Fusion

BM25 rank r_b

mDPR rank r_d

$$s_{\text{RRF}} = \frac{1}{k + r_b} + \frac{1}{k + r_d}$$

Final ranking